

SAULO QUIÑONES GÓNGORA

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[Portfolio](#) • [GitHub](#) • [LinkedIn](#) • [Certificates](#)

Professional Profile

Engineer working across three areas: web design and development, GIS and geospatial engineering, and applied AI. Trained as an Environmental Engineer, with a completed specialization in Artificial Intelligence (GPA 4.9/5.0) and an IBM Data Science Professional Certificate. Experience building interactive geospatial web tools, consolidating spatial databases from multiple institutional sources, and integrating AI into applied projects. Comfortable owning a project end to end, from data collection and backend logic to a working, shipped interface, turning messy real-world data into tools other people use.

Experience

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE (UAO) WITH ASOCAÑA

GIS Analyst and Environmental Engineer, Biological Corridor Project

Cali, Colombia

Feb 2026 – Present

Achievements and technical skills

- Led the collection and organization of environmental data for the Cauca River and its 15 priority tributaries, reconciling records from four regional environmental authorities (CVC, CARDER, CRC, IDEAM) that arrived with inconsistent formats, coordinate systems, and station names.
- Built the project's central geodatabase in ArcGIS Pro, consolidating water quality, streamflow, land use, and monitoring station layers under a single coordinate reference system to support all downstream spatial analysis.
- Designed and developed [Cauca River Baseline](#), a public GIS web viewer built with MapLibre GL JS that turns the geodatabase into an interactive tool for monitoring stations, water quality profiles, and streamflow duration curves, with CSV download per station. Used to share diagnostic results with the project's institutional and industry partners.
- Processed and standardized geospatial layers in ArcGIS Pro and Python (pandas, pyproj), including coordinate system normalization, attribute schema harmonization, and station name matching across datasets.
- Built and maintain a fully formula-linked Excel model, with no pasted values, that estimates diffuse pollutant load under three scenarios across ten interlinked sheets, so any input change propagates automatically.

ITESA, INSTITUTO TECNOLÓGICO SUPERIOR DE HIDALGO

Visiting Researcher, Programa Delfin 2025

Hidalgo, Mexico

Jun 2025 – Jul 2025

Achievements and technical skills

- Designed and built OptiCoder, a React Native and Expo mobile accessibility app integrating multimodal AI (Google Gemini for vision-language, OpenAI Whisper for speech-to-text) to help visually impaired developers interpret code screenshots, UI mockups, and architecture diagrams.
- Architected a layered system separating presentation, business logic, shared services, and remote AI infrastructure, with a two-stage prompting strategy for image analysis and voice interaction.
- Designed and ran an empirical evaluation with 7 software developers across 106 classification instances, reaching 74.1% precision and 58.1% recall in code-error detection, and revised the prompt design directly from evaluation findings, including a fix for the model misreading IDE error highlights instead of the code itself.
- Co-authored the resulting research paper, currently under review at an indexed journal in AI and accessibility.

FUDAMBIENT, CONTRACTOR FOR CVC

Environmental Professional, CVC Agreements N°031 and N°094

Cali, Colombia

Sep 2025 – Dec 2025

Achievements and technical skills

- Automated internal project workflows with Google Apps Script, including a system for organizing technical fact sheets on trees that replaced manual record keeping.
- Designed and developed FUDAMBIENT's institutional website using HTML, CSS, and JavaScript, structured around a clear presentation of services, projects, and results.
- Built small interactive web tools for environmental education (HTML, CSS, JavaScript), including an interactive map and drag-and-drop browser activities.
- Authored the environmental education plan (PRAE) for a public school in Caicedonia and supported rural wastewater treatment systems (SITAR).

ASOASIDE, CONTRACTOR FOR CVC

Environmental Professional, CVC Agreement N°004

Cali, Colombia

Mar 2025 – Sep 2025

Achievements and technical skills

- Supported socio-environmental diagnostics for river basins across the Valle del Cauca, applying participatory methods such as social cartography and problem trees to prioritize needs by municipality.
- Produced and reviewed technical documentation and tracking matrices for municipal environmental education plans (PMEA), keeping consistency with CVC guidelines.
- Built simple interactive web activities (HTML, CSS, JavaScript) to support environmental education materials for municipal processes.

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE (UAO)

Undergraduate Researcher, Semillero A+

Cali, Colombia

Jan 2024 – Nov 2024

Achievements and technical skills

- Built statistical and data mining models to assess the hydrobiological water quality of the Cali River for the undergraduate thesis, using biotic indices (BMWP/Col), habitat suitability models, and PCA in Python and XLSTAT.
- Analyzed raw air quality monitoring data (PM₁₀ and PM_{2.5}) from Cali's MIO public transit system, including exploratory and statistical analysis and mapping in Python (pandas, Matplotlib, Folium).
- Presented the air quality exposure analysis at RedCOLSI, scoring 94/100.

Selected Projects

- **EUDR Forest Risk Assessment.** Screens 4,170 real agricultural parcels for EU deforestation-free compliance using only open satellite data. Python, PostGIS, Google Earth Engine, scikit-learn, Docker. github.com/saulo1112/EUDR_Risk_Assessment
- **Cauca River Baseline.** Interactive GIS web viewer for river water quality and land use monitoring, built with MapLibre GL JS over a consolidated multi-source geodatabase. github.com/saulo1112/RioCauca_Baseline
- **OptiCoder.** Mobile multimodal AI accessibility app that reads code screenshots, UI mockups, and diagrams aloud for developers who cannot see them. React Native, Expo, Gemini, Whisper. github.com/saulo1112/PDproject
- **Site Selection Engine.** Spatial machine learning tool that scores candidate retail locations against neighborhood accessibility patterns using hexagonal grid analysis. Python, GIS. https://github.com/saulo1112/Site_selection_engine
- **App Physical Recovery.** Turns a phone camera into a physiotherapy coach using person detection and pose estimation. PyTorch, HuggingFace, gRPC, Streamlit, Docker. https://github.com/saulo1112/App_PhysicalRecovery

Education and Certifications

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE

Specialization in Artificial Intelligence

Cali, Colombia

Expected Oct 2026

- GPA 4.9/5.0. Coursework in machine learning, deep learning, computer vision, and applied AI systems.

B.Eng. in Environmental Engineering

Graduated May 2025

- GPA 4.7/5.0. Academic Excellence Award in semesters 2021-1, 2022-1, 2024-1, and 2024-3.

CERTIFICATIONS

- IBM Data Science Professional Certificate: Python, SQL, data analysis, and machine learning.
- EF SET English Certificate, level C2. Colegio Los Andes, high school valedictorian (2020).

Skills

Programming languages: Python, JavaScript, TypeScript, SQL, R, Google Apps Script.

Web and mobile: React Native, Expo, Astro, Node.js, FastAPI, Flask, MapLibre GL JS, Three.js, HTML, CSS.

GIS and geospatial: ArcGIS Pro (arcpy), QGIS, PostGIS, Google Earth Engine, pyproj, GeoJSON.

Machine learning and AI: PyTorch, scikit-learn, OpenCV, pandas, multimodal LLM integration (Gemini, Whisper).

Other tools: Git and GitHub, Docker, Power BI, XLSTAT, Qual2k, JupyterLab, advanced spreadsheet automation.

Languages: Spanish (native), English (C2).

Volunteering

UN VOLUNTEERS COLOMBIA

Logistics Assistant, COP16 Cali

Cali, Colombia

Oct 2024 – Nov 2024

- Coordinated participant guidance and transport logistics for a large-scale international summit, working bilingually with delegations.